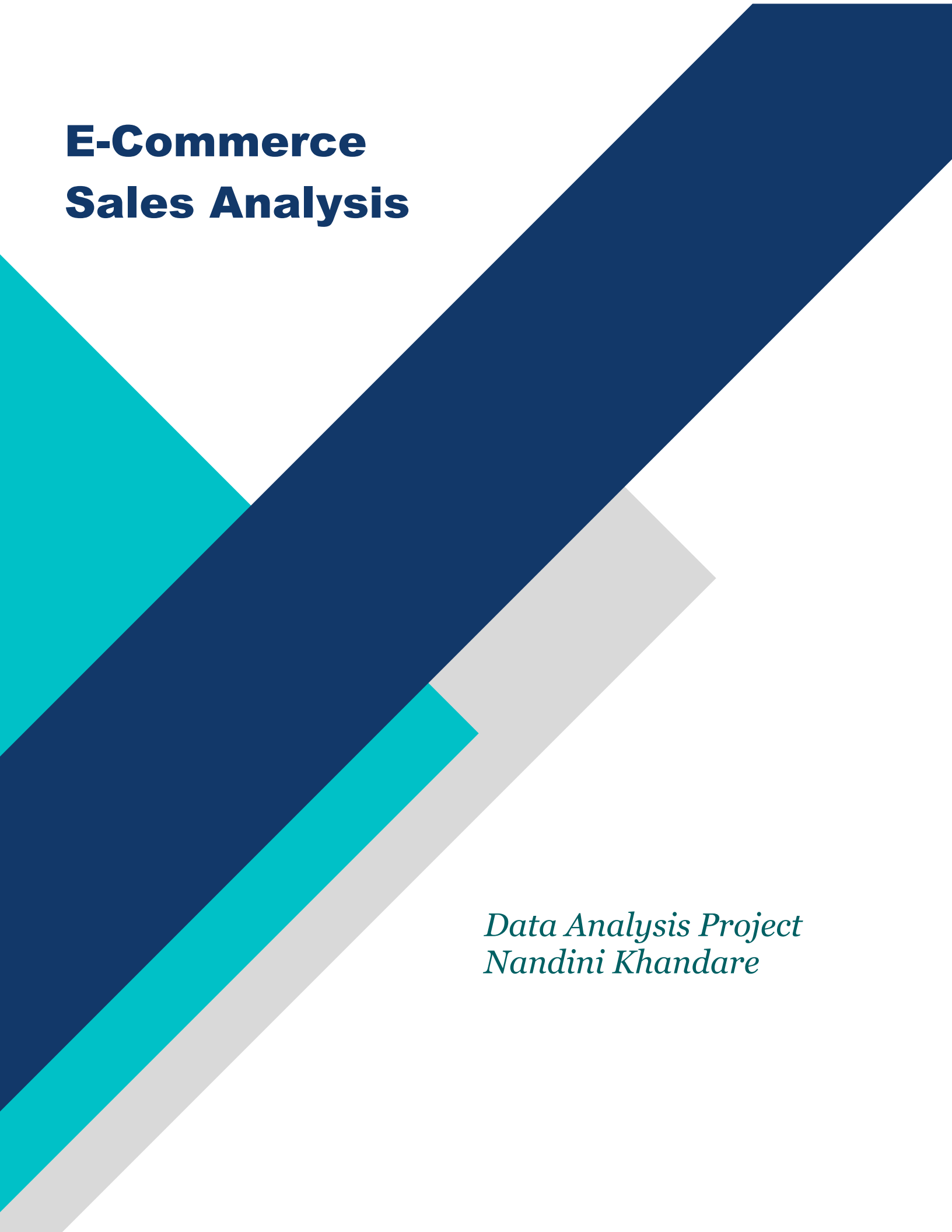




E-Commerce Sales Analysis

*Data Analysis Project
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Project Title :

Global E-Commerce Sales Analysis
using Excel, SQL Server, and Power BI

Task 1:

Data Cleaning Using Excel

- Removed duplicate records.
- Handled missing values.
- Converted Order Date and Ship Date to Date format.
- Removed blank rows and unnecessary columns.
- Checked numeric columns for consistency.
- Standardized text values (Category, Region, Market, Segment).
- Verified data types.

Excel Analysis:

- Calculate Total Sales
Result: ≈ 12.64 Million
- Calculate Total Profit
Result: ≈ 1.47 Million
- Find Average Discount
Result: $\approx 14.3\%$

- Top 5 Product by Sales

Created using:

- Pivot Table
- Product Name
- Sum of Sales
- Sort Descending

- Top 5 loss-Making Products

Created using:

- Pivot Table
- Product Name
- Sum of Sales
- Sort Ascending

- Region-wise Sales

Created using Pivot Table.

Pivot Tables and Charts

1. Sales by region chart

Objective

To identify which regions contribute the highest and lowest sales.

Analysis

- The chart compares total sales across all regions.
- Regions with higher sales are the company's strongest markets and contribute significantly to overall revenue.
- Regions with lower sales may indicate weaker customer demand, limited market penetration, or fewer marketing efforts.

Business Insight

- Focus on maintaining growth in the highest-performing regions.
- Increase marketing campaigns and promotional offers in low-performing regions to improve sales.

2. Profit by Category Chart

Objective

To compare profitability across product categories.

Analysis

- The chart shows the total profit earned from each product category.
- Categories generating higher profits are more valuable for the business.
- Categories with lower profit may have high discounts, higher shipping costs, or lower sales margins.

Business Insight

- Continue investing in high-profit categories.
- Review pricing, discount strategies, and operational costs for low-profit categories to improve profitability.

3. Segment-wise Sales Analysis

Objective

To analyze sales performance across different customer segments.

Analysis

- The chart displays sales generated by Consumer, Corporate, and Home Office segments.
- The segment with the highest sales represents the largest customer base.
- Lower-performing segments may offer opportunities for business expansion.

Business Insight

- Develop loyalty programs for the highest revenue-generating customer segment.
- Introduce targeted marketing strategies to increase sales in lower-performing segments.

4. Monthly Sales Trend

Objective

To analyze sales performance over time.

Analysis

- The line chart illustrates monthly changes in total sales.
- Sales fluctuate throughout the year, showing periods of both growth and decline.
- Higher sales in the final months of the year indicate seasonal demand and increased customer purchasing.

Business Insight

- Prepare inventory and staffing for peak sales months.
- Launch promotional campaigns during low-sales months to improve overall revenue.

Task 2:

SQL Analysis

1. Top 10 profitable Product

Objective

Identify the products that generate the highest total profit.

SQL Query

```
SELECT TOP 10
    product_name,
    SUM(profit) AS Total_Profit
FROM ecommerce_cleaned_data
GROUP BY product_name
ORDER BY Total_Profit DESC;
```

Analysis

The query groups all orders by product and calculates the total profit earned by each product.

Business Insight

- High-profit products contribute significantly to business growth.
- These products should receive priority in inventory planning and promotional campaigns.
- Maintaining stock availability for these products can increase overall profitability..

2. Top 10 customers by Sales

Objective

Identify customers who contribute the highest sales revenue.

SQL Query

```
SELECT TOP 10
    customer_name,
    SUM(sales) AS Total_Sales
FROM ecommerce_cleaned_data
GROUP BY customer_name
```

ORDER BY Total_Sales DESC;

Analysis

The query calculates the total sales generated by each customer and ranks them in Descending order.

Business Insight

- A small number of customers generate a large portion of total sales.
- Loyalty programs and personalized offers can help retain these valuable customers.
- Customer relationship management should focus on these high-value clients.

3.Region-wise Total Sales

Objective

Analyze sales performance across different regions.

SQL Query

```
SELECT
    region,
    SUM(sales) AS Total_Sales
FROM ecommerce_cleaned_data
GROUP BY region
ORDER BY Total_Sales DESC;
```

Analysis

The query sums sales for each region to identify the highest and lowest performing Market.

Business Insight

- High-performing regions generate the majority of revenue.
- Low-performing regions may require targeted marketing and business expansion strategies.

4.Category-wise Average Sales

Objective

Compare the average profit generated by each product category.

SQL Query

```
SELECT
    category,
    AVG(profit) AS Average_Profit
FROM ecommerce_cleaned_data
GROUP BY category
ORDER BY Average_Profit DESC;
```

Analysis

The query calculates the average profit earned per order for each product category.

Business Insight

- Categories with higher average profit provide better returns.
- Low-profit categories should be reviewed for pricing, discounts, and operational costs.

5.Highest Discount Category

Objective

Identify which category receives the highest average discount.

SQL Query

```
SELECT
    category,
    AVG(discount) AS Average_Discount
FROM ecommerce_cleaned_data
```

GROUP BY category
ORDER BY Average_Discount DESC;

Analysis

The query computes the average discount offered within each category.

Business Insight

- Categories with higher discounts may attract more customers but can reduce profitability.
- Discount policies should be optimized to balance sales growth and profit margins.

6.Order with Negative Profit

Objective

Find all orders where the company incurred a loss.

SQL Query

```
SELECT
    order_id,
    product_name,
    sales,
    profit
FROM ecommerce_cleaned_data
WHERE profit < 0;
```

Analysis

The query filters orders where profit is less than zero.

Business Insight

- Loss-making orders should be investigated.
- High discounts, shipping costs, or pricing issues may be responsible.
- Reducing such losses can improve overall business performanceDiscount policies should be optimized to balance sales growth and profit margins.

7.Monthly Sales Trend

Objective

Analyze monthly sales performance.

SQL Query

```
SELECT
    YEAR(order_date) AS Year,
    MONTH(order_date) AS Month,
    SUM(sales) AS Total_Sales
FROM ecommerce_cleaned_data
GROUP BY YEAR(order_date), MONTH(order_date)
ORDER BY Year, Month;
```

Analysis

The query groups sales by year and month to identify sales trends over time..

Business Insight

- Sales fluctuate throughout the year.
- Peak months require sufficient inventory and workforce planning.
- Marketing campaigns can be scheduled during slower months to improve revenue.

8.Market- wise Revenue Analysis

Objective

Compare revenue generated across different markets.

SQL Query

```
SELECT
```

```
market,  
SUM(sales) AS Revenue  
FROM ecommerce_cleaned_data  
GROUP BY market  
ORDER BY Revenue DESC;
```

Analysis

The query calculates total revenue for each market.

Business Insight

- High-performing markets should remain a strategic focus.
- Lower-performing markets may offer opportunities for expansion through localized marketing efforts.

9.Top Performing Sub-Categories

Objective

Identify the best-performing product sub-categories based on sales.

SQL Query

```
SELECT  
    sub_category,  
    SUM(sales) AS Total_Sales  
FROM ecommerce_cleaned_data  
GROUP BY sub_category  
ORDER BY Total_Sales DESC;
```

Analysis

The query ranks sub-categories according to total sales.

Business Insight

- High-performing sub-categories should receive greater inventory allocation and marketing support.
- Understanding customer preferences helps improve product planning.

10.Ship Mode Usage Analysis

Objective

Determine which shipping method is used most frequently.

SQL Query

```
SELECT  
    ship_mode,  
    COUNT(*) AS Total_Orders  
FROM ecommerce_cleaned_data  
GROUP BY ship_mode  
ORDER BY Total_Orders DESC;
```

Analysis

The query counts the number of orders shipped through each shipping mode.

Business Insight

- Standard Class is typically the most commonly used shipping method because it balances cost and delivery time.
- Businesses can optimize logistics by focusing on the most frequently used shipping methods while offering premium shipping options for urgent deliveries

Task 3:

Power BI Dashboard

KPI Cards

- Total Sales
- Total Profit
- Total Orders
- Average Discount

Charts

- Sales by Region
- Sales by Market
- Monthly Sales Trend
- Category Performance
- Top Products
- Top Customers
- Profit by Sub-Category
- Segment Analysis

Slicers

- Year
- Region
- Market
- Segment
- Category

Business Insights

Sales Performance

- Total Sales exceeded **12.6 Million**, showing strong business performance.
- Total Profit reached approximately **1.47 Million**, indicating overall profitability.

Regional Insights

- The **Central** region generated the highest sales.
- Lower-performing regions may require targeted marketing and promotional strategies.

Category Performance

- **Technology** generated the highest sales.
- **Furniture** had comparatively lower profitability because of higher discounts and shipping costs.

Customer Insights

- A small group of customers contributed a significant portion of total sales.
- Rewarding these customers through loyalty programs could increase repeat purchases.

Product Performance

- Some products consistently generated losses.

- These products should be reviewed for pricing, procurement costs, and discount policies.

Discount Analysis

- Average discount was approximately **14.3%**.
- Higher discounts did not always result in higher profits.
- Discount strategies should be optimized to protect profit margins.

Shipping Analysis

- Standard Class was the most frequently used shipping mode.
- Faster shipping modes were used less often because of their higher cost.

Market Analysis

- One market contributed the highest revenue and should remain a strategic focus for future growth.
- Markets with lower sales present opportunities for expansion through localized campaigns.

Monthly Trend

- Sales increased toward the end of the year, indicating seasonal demand.
- Inventory and staffing should be planned accordingly for peak months.

Conclusion

The analysis demonstrates how Excel, SQL Server, and Power BI can be combined to transform raw transactional data into actionable business insights. Through data cleaning, SQL-based querying, and interactive dashboard visualization, the project identified high-performing markets, profitable product categories, customer trends, and areas requiring improvement. These insights can support better pricing decisions, marketing strategies, inventory planning, and overall business growth.

Thank You!!

